

Longitudinal Data Analysis of Housing Supply and Poverty Exposure in the U.S.

by Adrian Lopez. MATH 120, Fall 2025.

Introduction

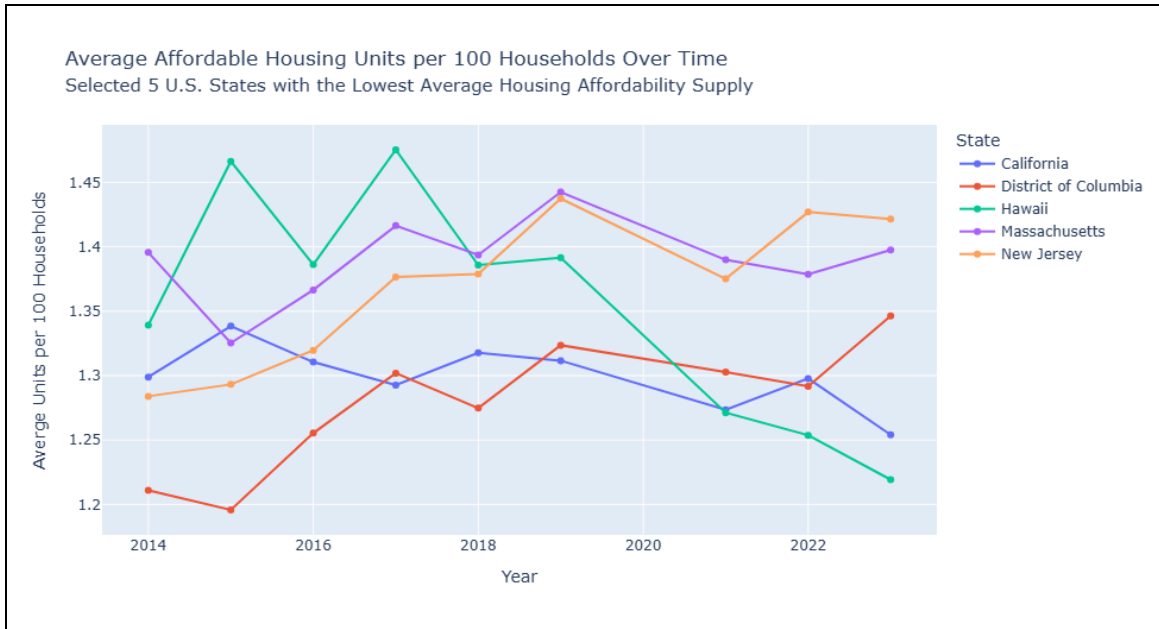
Considering the United States exists within a hierarchical system involving differences in class and status, my project focuses on understanding what factors contribute to moving up the ladder within this system. This idea is known as upward mobility which the Urban Institute defines its necessities as economic success, power, autonomy, dignity, and belonging. To identify which states in the U.S. lack these necessities, I used mobility metrics from the Urban Institute to address how the lowest average affordable housing supply and share of poverty exposure has changed from years 2014 to 2023 for U.S. states. To build off this idea, I determined whether states with more affordable housing shortages are associated with higher poverty exposure and higher homeless public-school students using linear regression. Ultimately, the results from this project can be used to inform which states should allocate resources for affordable housing and homelessness-prevention for students in public-schools across the nation.

Methods

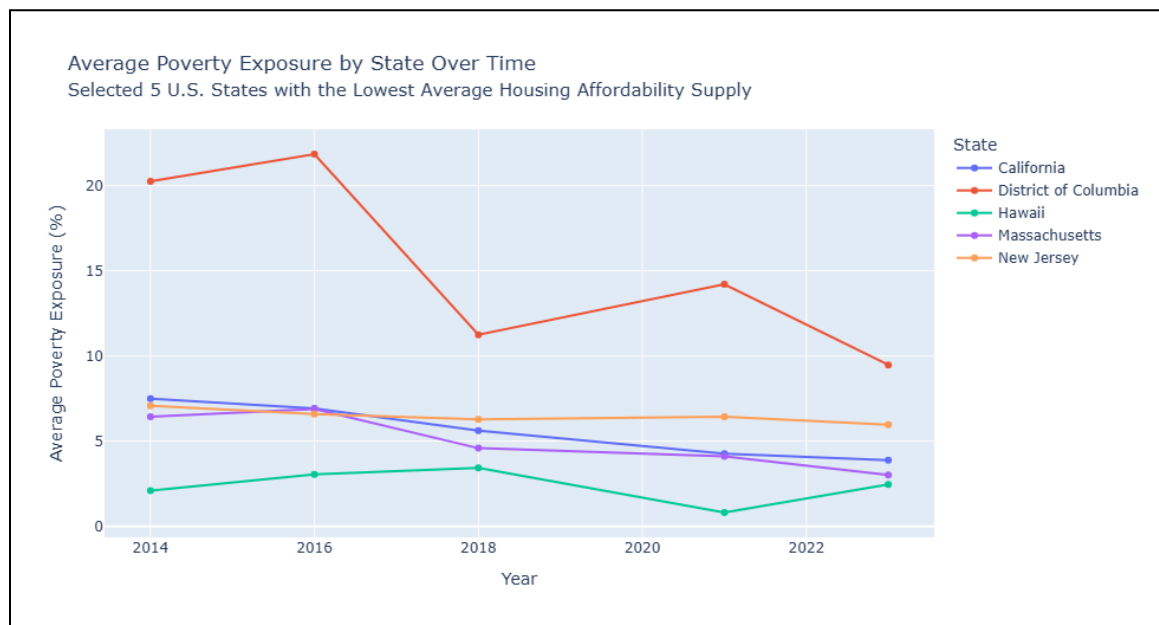
The county-level longitudinal mobility metrics used in this analysis are defined as the number of affordable housing units for every 100 households that are below 80% of the area median income, share of people experiencing poverty who live in high-poverty neighborhoods, and share of public-school children that are ever homeless during the school year. Prior to analysis, I cleaned the columns involving these variables by dropping missing values and acknowledged that the missing data for the year of 2020 was due to COVID-19 affecting data collection. To interpret these variables more easily, I converted these share values to percentages and aggregated these values to averages. To narrow my focus on these variables, I also selected five U.S. states with the lowest average affordable housing availability to use these states as a comparison group for analysis.

The visualizations used involved line charts that looked at average housing affordability and poverty exposure over time for these five states. To better see the relationship of these trends, I also plotted trendlines for average affordability versus poverty exposure and versus student homelessness. To analyze the strength of this relationship, I built linear regression models that used average affordability as a predictor variable of average poverty exposure and average percentage of student homelessness as a response variable, separately. The p-value and R-squared value were used to evaluate performance in these models on whether the relationship was statistically significant and how strong it was.

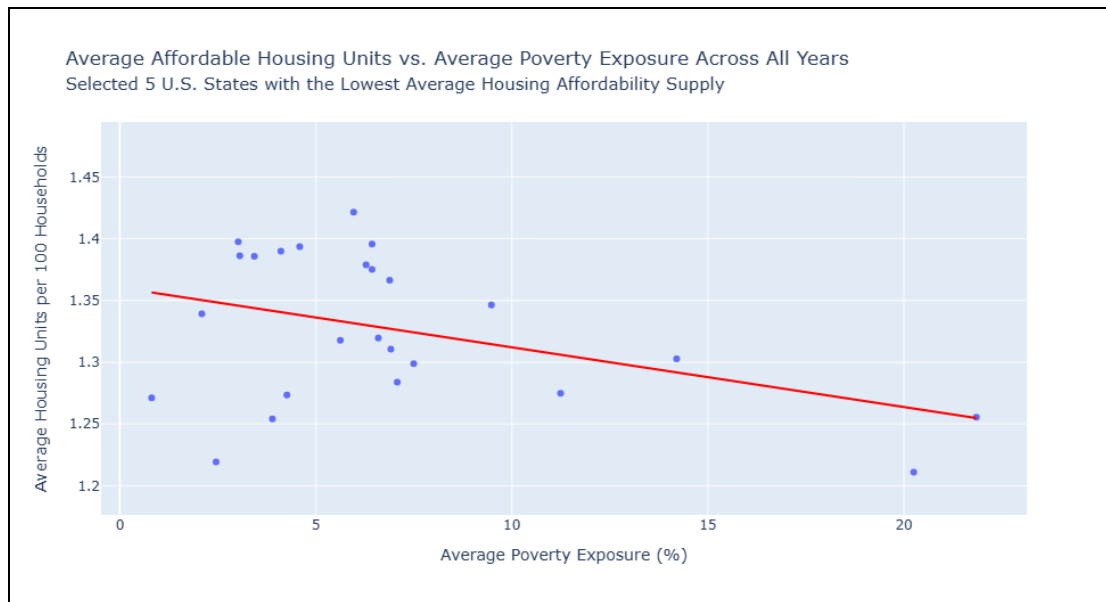
Results



The first figure above looks at five states in the U.S. that have the lowest average share of affordable housing units per 100 households that are specifically below 80% of the area median income and how it's changed over time. As you can see from the above graph, Hawaii had the highest peak in average share of affordable housing units per 100 households during the year of 2017. However, the District of Columbia had the lowest dip out of all 5 states in 2015.

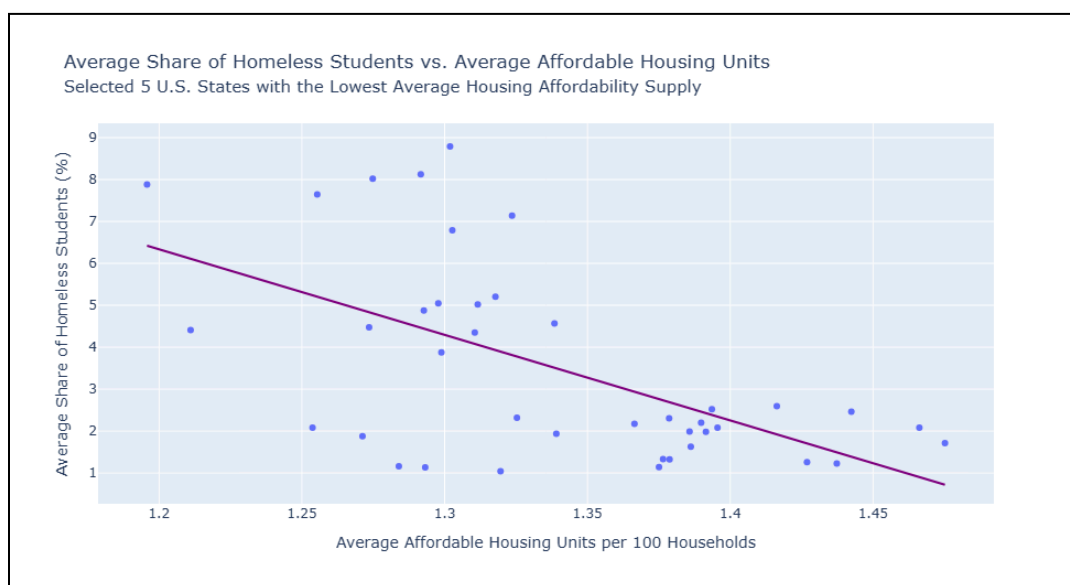


This next figure uses the same 5 states with the lowest average affordable housing units from the previous figure and plots those states by average poverty exposure. From the above graph, we can see that the District of Columbia has the highest peak out of all 5 states in average poverty exposure in 2016 while Hawaii has the lowest.



The figure above plots the average number of affordable housing units per 100 households versus average poverty exposure. As we can see, the graph shows that as average poverty exposure increases, the average number of affordable housing units decreases, exemplifying a negative association by the red linear regression line.

I explored the strength of this association in a linear model that used average housing affordability as the predictor variable and average poverty exposure as the response variable. The summary results show that the p-value was less than 0.05 for the average affordability variable which lets us know that there is a statistically significant linear association between average affordability and average poverty exposure. The R-squared value was 0.167 which means that affordability alone explains 16.7% of the variation in poverty exposure across these observations.



This last figure plots the average percentage of public-school children who are ever homeless during the school year versus the average number of affordable housing units per 100 households. As you can see from the graph, as the average housing affordability increases, the percentage of homeless students decreases which suggests a negative association between the two.

I further explored this association in a linear regression model that used average housing affordability as the predictor variable and average share of homeless students as the response variable. The summary results for this model were similar to the previous one and showed that there is a statistically significant linear association between average affordability and the average share of homeless students. The R-squared value for this model was 0.33 which means that affordability explains 33% of the variation in student homelessness across these observations.

Discussion

The results showed that for the selected five U.S. states with the lowest average housing affordability, housing affordability is linked to both poverty exposure and student homelessness. This suggests that when the supply of affordable housing is low, families may be more likely to live where poverty is more concentrated. It also suggests that homelessness for children in public schools is also likely due to that lack of affordable housing.

It should be noted that there are many other factors that influence poverty exposure and student homelessness which is part of the limitations of this analysis. If I had more time, I would build regression models that take into account more mobility metrics to better understand more factors that influence these two variables.

Conclusions

Overall, this analysis used Urban Institute county-level longitudinal mobility metrics to focus on five U.S. states with the lowest average affordable housing availability. The key findings showed that average affordable housing availability is negatively associated with poverty exposure and student homelessness. These patterns can inform policy by showing that the investment of affordable housing resources and student homelessness-prevention should be more allocated towards the states of California, District of Columbia, Hawaii, Massachusetts, and New Jersey.

References

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